Real-Parameter Heavy-Ball Dynamics on Every SPD Spectral Interval: Exact Jury Triangle, Sharp Root Factor, and Jordan Transients

Route-A structural certificate C201

Abstract

We give one complete phase atlas for constant-parameter Polyak heavy-ball dynamics on every real SPD quadratic and for all real step and momentum parameters. Scalar modal reduction yields the exact Jury triangle, including negative momentum, and the exact endpoint root radius. A disk-Jury argument proves the unique minimax Polyak pair. Its endpoint blocks are defective, so the sharp root factor q comes with generic $O(kq^k)$ transients rather than a uniform Cq^k bound. We also close the equal-spectrum nilpotent case, full characteristic data, conformal symplectic identity and every finite-order boundary. Exact independent certificates test the formulas but do not prove the continuum theorem. No claim extends to arbitrary nonlinear objectives or to target arithmetic, and Route A is strictly rejected.

Keywords: heavy-ball method; Polyak momentum; Jury criterion; quadratic optimization; spectral radius; Jordan block; symplectic map.

中文摘要

本文对全部实步长与实动量参数下的正定二次型重球迭代给出完整相图。 模态分解导出包含负动量的精确Jury稳定三角；圆盘判据证明唯一极小极大 Polyak参数。最优端点均为缺陷Jordan块， 因此精确根收敛因子为q，但一般瞬态是$O(kq^k)$而非一致Cq^k。 本文不把二次型结论推广到一般非线性强凸函数，也不提出目标算术或路线B结论。
关键词：重球法；稳定区域；谱半径；缺陷矩阵；辛边界。

1 Modal owner and exact Jury triangle

Let

$$f(x) = \tfrac{1}{2}x^T Ax - b^T x, \qquad 0 < mI \le A \le LI,$$

and let $x_* = A^{-1}b$. Polyak's iteration [1] gives

$$e_{k+1} = [(1+\beta)I - \alpha A]e_k - \beta e_{k-1}, \qquad e_k = x_k - x_*. \tag{1}$$

For an eigenvalue λ, the scalar characteristic polynomial is

$$p_\lambda(r) = r^2 - a_\lambda r + \beta, \qquad a_\lambda = 1 + \beta - \alpha\lambda. \tag{2}$$

For a real monic quadratic, Jury's criterion is $|\beta| < 1$, $p_\lambda(1) > 0$ and $p_\lambda(-1) > 0$. Here

$$p_\lambda(1) = \alpha\lambda, \qquad p_\lambda(-1) = 2(1+\beta) - \alpha\lambda.$$

Therefore the whole spectral class converges for every phase initialization if and only if

$$\boxed{-1 < \beta < 1, \qquad 0 < \alpha < 2(1+\beta)/L.} \tag{3}$$

This is the complete real-parameter region, not merely the usual $0 \le \beta < 1$ sector.

For $a \in \mathbb{R}$, the exact root radius is

$$R_\beta(a) = \begin{cases} \sqrt{\beta}, & \beta \ge 0,\ |a| \le 2\sqrt{\beta}, \\ (|a| + \sqrt{a^2 - 4\beta})/2, & \text{otherwise.} \end{cases} \tag{4}$$

It is nondecreasing in $|a|$ outside the plateau. Since a_λ is affine, the robust radius is exactly $\max\{R_\beta(a_m), R_\beta(a_L)\}$.

Revision focus. Round 0 closes the modal family, exact all-real Jury triangle and endpoint radius.

Declarations

Data and code availability. Package-local synthetic exact evidence and deterministic code accompany the manuscript.

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